Package 'levyfit'

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Title Maximum Likelihood Estimation for Levy Degradation Processes

Version 0.1.0

Description Maximum likelihood estimation of the parameters of degradation models based on Levy processes (Wiener, gamma, inverse Gaussian, variance-gamma and Meixner processes) from one or several trajectories observed at regular or irregular inspection times.

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URL https://github.com/rytracetito/levy_packages/tree/main/levyfit, https://github.com/rytracetito/levy_packages

BugReports https://github.com/rytracetito/levy_packages/issues

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Contents

levyfit-package 2
gamma_mle 5
ig_mle 8
increments_data 11
meixner_mle 12
vg_mle 17
wiener_mle 22

Index **26**

levyfit-package *levyfit: maximum likelihood estimation for degradation processes*

Description

levyfit estimates the parameters of stochastic degradation models from one or several degradation trajectories observed at regular or irregular inspection times. All the parameters are estimated by maximum likelihood (maximum likelihood estimator, MLE). The help page of each function gives the model, the likelihood and how the estimator is computed.

Functions of the package

Estimation of the parameters:

Function	Process	Parameters	Computation
wiener_mle()	Wiener with drift	mu, sigma2, s2	closed form
gamma_mle()	gamma	shape a, rate b	root of one equation
ig_mle()	inverse Gaussian	mu, eta, eta_unbiased	closed form
vg_mle()	variance-gamma	theta, sigma, nu	EM, then L-BFGS-B
meixner_mle()	Meixner	a, b, d, m	profile likelihood

s2 and eta_unbiased are the unbiased versions of sigma2 and eta. Every estimation function returns the point estimates and, with ci = TRUE, asymptotic confidence intervals. increments_data() gives the data by their increments instead of their levels.

Inspection times following a chosen pattern, and simulated trajectories to try the estimators, are given by the package DegradSim (DegradSim::inspection_times() and its simulation functions).

Data

Every estimation function accepts the data in four forms (shown here with gamma_mle(); the call is the same for all the estimation functions):

- one trajectory: gamma_mle(x, t), with x the degradation levels and t the inspection times;
- several trajectories with the same inspection times: gamma_mle(x), with x the matrix returned by the DegradSim simulation functions (columns t, X1, X2, ...);
- several trajectories, each with its own inspection times: gamma_mle(liste), with liste a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels;
- the increments themselves, given with the function increments_data().

The trajectories are assumed independent and with the same distribution. The increments are computed within each trajectory and then pooled.

Notation

The same notation is used in all the help pages of the package.

- $(\Delta t_j, \Delta x_j)$, $j = 1, \ldots, n$: the durations and the degradation increments between two consecutive inspections of a same trajectory, pooled over all the trajectories, so that n is the total number of increments. The durations Δt_j can be equal (regular inspections) or not: all the formulas hold in both cases.
- $T = \sum_{j=1}^{n} \Delta t_j$, the total observed time, and $S = \sum_{j=1}^{n} \Delta x_j$, the total degradation.
- θ: the vector of the parameters of the Levy process, for instance $\theta = (a, b)$ for the gamma process.
- $f_{\Delta t}(x; \theta)$: the density of an increment of the process over a duration Δt.
- ψ and ψ': the digamma and trigamma functions, the first and second derivatives of $\ln \Gamma$.

Computation of the estimators

The increments of a Levy process being independent, the log-likelihood is

$$\ell(\theta) = \sum_{j=1}^{n} \ln f_{\Delta t_j}(\Delta x_j; \theta).$$

- **Closed form**: Wiener and inverse Gaussian. The MLE is explicit, with $\hat{\mu} = S/T$.
- **Root of one equation**: gamma. The estimator $\hat{b} = \hat{a}T/S$ is explicit and $\hat{a}$ is the unique root of a decreasing function, found by `stats::uniroot()` from the method of moments estimator; the result does not depend on this starting point.
- **Numerical maximization**: variance-gamma and Meixner.
  - Variance-gamma: the process is a Brownian motion observed at a random time given by a gamma process. The EM algorithm treats this random time as a hidden variable (each step is explicit or a unique root); it starts from the Wiener MLE and is followed by L-BFGS-B (`stats::optim()`).
  - Meixner: L-BFGS-B (`stats::optim()`) on the likelihood profiled in b ($\hat{b}$ is explicit for given a, d, m), with the exact gradient, from several starting points: the moment estimates when they exist, and points built from the empirical mean and variance (`n_start`); the solution of largest likelihood is kept.

Confidence intervals

With `ci = TRUE`, the intervals are asymptotic intervals based on the asymptotic normality of the MLE. For i.i.d. observations, that is equal durations, the MLE is asymptotically normal with covariance matrix the inverse of the Fisher information (van der Vaart, 1998, Theorem 5.39); with unequal durations the increments are independent but not identically distributed, and the same normal approximation is used. Let $\hat{\theta}$ be the MLE. The Fisher information is estimated by the observed information $J = -\partial^2 \ell / \partial\theta \partial\theta^\top$ taken at $\hat{\theta}$ (Grigoletto and Provasi, 2009, section 4.2), computed numerically by `stats::optimHess()`, except for the Meixner process whose Hessian is explicit (see `meixner_mle()`). The standard error of the k-th parameter is $se_k = \sqrt{(J^{-1})_{kk}}$; replacing the unknown standard deviation by a consistent estimate keeps the asymptotic level of the interval (van der Vaart, 1998, section 3.2). Let $z = \Phi^{-1}((1 + level)/2)$, where Φ is the distribution function of the standard normal distribution: $(1 + level)/2$ is the order of the quantile for a two-sided interval, for instance $z = 1.96$ for `level = 0.95`. The interval is

- $\hat{\theta}_k \pm z\, se_k$ for a real parameter (`mu` of the Wiener process, `theta` of the variance-gamma process, `b` and `m` of the Meixner process);
- $[\hat{\theta}_k e^{-z\, se_k/\hat{\theta}_k},\ \hat{\theta}_k e^{z\, se_k/\hat{\theta}_k}]$ for a positive parameter (all the others): this is the interval $\ln \hat{\theta}_k \pm z\, se_k/\hat{\theta}_k$ for $\ln \theta_k$, whose standard error $se_k/\hat{\theta}_k$ is given by the delta method (van der Vaart, 1998, Theorem 3.1); since $\ln$ is monotone, this interval is transformed back into an interval for θ_k (van der Vaart, 1998, section 3.2), which always stays positive.

The unbiased estimates `s2` (Wiener) and `eta_unbiased` (inverse Gaussian) are given without interval. When J is not positive definite, the bounds are `NA`.

For the Wiener process, the information matrix of Kahle, Mercier and Paroissin (2016, equation [1.17]), written for the n increments, gives $se(\hat{\mu}) = \hat{\sigma}/\sqrt{T}$ and $se(\hat{\sigma}^2) = \hat{\sigma}^2\sqrt{2/n}$.

Parametrizations

- Wiener: $X_t - X_s \sim \mathcal{N}(\mu(t-s), \sigma^2(t-s))$ (Kahle et al., 2016, definition 1.11).
- Gamma: $X_t - X_s \sim \Gamma(a(t-s), b)$, with shape $a(t-s)$ and rate b (scale $1/b$), so that $E[X_t] = at/b$ (Kahle et al., 2016, definition 2.24 and section 2.7).
- Inverse Gaussian: $X_t - X_s \sim IG(\mu(t-s), \eta(t-s)^2)$, the process of Wang and Xu (2010) with $\Lambda(t) = \mu t$ (see `ig_mle()` for the correspondence of the parameters).
- Variance-gamma: $X_s = \theta\gamma_s + \sigma W_{\gamma_s}$, with γ_s of mean s and variance νs (Seneta, 2004; Bee, Dickson and Santi, 2018).
- Meixner: $X_s \sim Meixner(a, b, ds, ms)$, the Levy process associated with the Meixner distribution of Grigoletto and Provasi (2009) (see `meixner_mle()`).

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References

Degradation processes: Kahle, W., Mercier, S. and Paroissin, C. (2016). *Degradation Processes in Reliability*. ISTE and Wiley, London. Chapter 1 (Wiener process) and chapter 2 (gamma process, in particular section 2.7 on the maximum likelihood and moment estimators).

Gamma process: van Noortwijk, J. M. (2009). A survey of the application of gamma processes in maintenance. *Reliability Engineering & System Safety*, 94(1), 2-21. doi:10.1016/j.ress.2007.03.019. Section 3.7.1 (maximum likelihood estimation).

Wiener process: Whitmore, G. A. (1995). Estimating degradation by a Wiener diffusion process subject to measurement error. *Lifetime Data Analysis*, 1, 307-319.

Inverse Gaussian process: Wang, X. and Xu, D. (2010). An inverse Gaussian process model for degradation data. *Technometrics*, 52(2), 188-197. doi:10.1198/TECH.2009.08197

Variance-gamma process: Seneta, E. (2004). Fitting the variance-gamma model to financial data. *Journal of Applied Probability*, 41A, 177-187. doi:10.1017/S0021900200112288. Bee, M., Dickson, M. M. and Santi, F. (2018). Likelihood-based risk estimation for variance-gamma models. *Statistical Methods & Applications*, 27(1), 69-89. doi:10.1007/s102600170393z

Meixner process: Grigoletto, M. and Provasi, C. (2009). Simulation and estimation of the Meixner distribution. *Communications in Statistics - Simulation and Computation*, 38(1), 58-77. doi:10.1080/03610910802395679

Confidence intervals: van der Vaart, A. W. (1998). *Asymptotic Statistics*. Cambridge University Press. doi:10.1017/CBO9780511802256. Theorem 3.1, section 3.2 and Theorem 5.39.

The references of each method are given in the help page of each function.

See Also

Useful links:

- https://github.com/rytracetito/levy_packages/tree/main/levyfit
- https://github.com/rytracetito/levy_packages
- Report bugs at https://github.com/rytracetito/levy_packages/issues

Examples

```
# 1. Inspection times: 10 trajectories inspected more and more often
plan <- DegradSim::inspection_times("decreasing", n = 20, n_traj = 10, tmax = 100)

# 2. Simulated data, here a gamma process (a = 2, b = 3)
set.seed(1)
x <- DegradSim::sim_gamma(plan, a = 2, b = 3)

# 3. Estimation of several models on the same data
gamma_mle(x)
wiener_mle(x)
ig_mle(x)

# 4. Estimates with 95 % confidence intervals
gamma_mle(x, ci = TRUE)
```

`gamma_mle` *Maximum likelihood estimation for the gamma process*

Description

Computes the maximum likelihood estimator (MLE) of the shape a and the rate b of the gamma process from one or several trajectories observed at regular or irregular inspection times. The MLE always exists (with probability one), is unique, and is the root of an equation in one variable.

Usage

```
gamma_mle(x, t = NULL, ci = FALSE, level = 0.95)
```

Arguments

`x`	Degradation levels of one trajectory, with strictly positive increments; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels. The increments can also be given directly, with the function `increments_data()`.
`t`	Inspection times of the trajectory, strictly increasing. Not used when `x` is a list.
`ci`	`FALSE` (default) for the point estimates only; `TRUE` to add asymptotic confidence intervals, computed as explained in the section Confidence intervals of levyfit-package.
`level`	Confidence level of the intervals, between 0 and 1 (default 0.95). Used only when `ci = TRUE`.

Value

A named vector with `a` $= \hat{a}$ and `b` $= \hat{b}$.

With `ci = TRUE`, a data frame with one row per parameter and the columns `parameter`, `estimate`, `lower` and `upper` (bounds of the confidence interval). When the estimator does not exist, the estimates (and bounds) are `NA`, with a warning.

Model and data

The gamma process is increasing, with independent increments $X_t - X_s \sim \Gamma(a(t-s), b)$, with shape $a(t-s)$ and rate b (Kahle, Mercier and Paroissin, 2016, definition 2.24 with $A(t) = at$), so that $E[X_t] = at/b$ and $Var(X_t) = at/b^2$ (Kahle et al., 2016, section 2.7). The density of an increment over a duration Δt is (Kahle et al., 2016, definition 2.1)

$$f_{\Delta t}(x; a, b) = \frac{b^{a\Delta t}}{\Gamma(a\Delta t)}\, x^{a\Delta t - 1} e^{-bx}, \qquad x > 0.$$

The data are the n increments $(\Delta t_j, \Delta x_j)$, $j = 1, \ldots, n$, with $\Delta x_j > 0$, pooled over the trajectories, with $T = \sum_{j=1}^n \Delta t_j$ and $S = \sum_{j=1}^n \Delta x_j$ (see the section Notation of levyfit-package). Let also

$$U = \sum_{j=1}^n \Delta t_j \ln \Delta x_j, \qquad r_j = \frac{\Delta x_j}{\Delta t_j} \text{ (speeds)}, \qquad w_j = \frac{\Delta t_j}{T} \text{ (weights)}.$$

Likelihood

The increments being independent, the log-likelihood is (Kahle et al., 2016, equation [2.42] with $\Delta A_{i,j}(\theta) = a s_{i,j}$)

$$\ell(a, b) = \sum_{j=1}^n \left[(a\Delta t_j - 1)\ln \Delta x_j - \ln \Gamma(a\Delta t_j) + a\Delta t_j \ln b - b\Delta x_j\right].$$

It is a two-parameter exponential family with sufficient statistic (U, S), whatever the durations Δt_j, equal or not. With equal durations Δt, the increments are i.i.d. $\Gamma(a\Delta t, b)$ and (U, S) is, up to the factor Δt, the usual sufficient statistic $(\sum_{j=1}^n \ln \Delta x_j, \sum_{j=1}^n \Delta x_j)$ of an i.i.d. gamma sample.

Maximum likelihood estimator

The score in b is $\partial \ell / \partial b = aT/b - S$, so that, for a fixed a, ℓ is maximal at $\hat{b}(a) = aT/S$. The profile log-likelihood $\ell_p(a) = \ell(a, \hat{b}(a))$ has derivative

$$g(a) = U - \sum_{j=1}^n \Delta t_j\, \psi(a\Delta t_j) + T \ln \frac{aT}{S},$$

where ψ is the digamma function. The function $x \mapsto \ln x - \psi(x)$ is strictly completely monotonic on $(0, \infty)$ (Alzer, 1997, Theorem 1 with $\alpha = 0$), hence strictly decreasing, so that $\psi'(x) > 1/x$ for $x > 0$; therefore $g'(a) = -\sum_{j=1}^n \Delta t_j^2 \psi'(a\Delta t_j) + T/a < 0$ and g is continuous and strictly decreasing on $(0, \infty)$. Moreover, by the inequality $1/(2x) < \ln x - \psi(x) < 1/x$, $x > 0$ (Alzer, 1997, equation (2.2)), $g(a) > g_0 + n/(2a)$ with g_0 not depending on a, so that $g(a) \to +\infty$ as $a \to 0$, and $\psi(x) - \ln x \to 0$ as $x \to \infty$, so that, by Jensen's inequality,

$$\lim_{a \to \infty} g(a) = T\Big(\sum_{j=1}^n w_j \ln r_j - \ln \sum_{j=1}^n w_j r_j\Big) < 0,$$

unless all the speeds r_j are equal. By the intermediate value theorem, g has a root $\hat{a}$, unique since g is strictly decreasing, and $\hat{b} = \hat{a}T/S$. These are the equations [2.43] and [2.44] of Kahle, Mercier and Paroissin (2016, section 2.7.1.1). With equal durations Δt, the equation $g(a) = 0$ becomes equation [2.45a] of Kahle et al. (2016), whose common step δ is Δt here:

$$\psi(a\Delta t) - \ln(a\Delta t) = \frac{1}{n}\sum_{j=1}^{n} \ln \Delta x_j - \ln\Big(\frac{1}{n}\sum_{j=1}^{n} \Delta x_j\Big),$$

the logarithm of the ratio of the geometric and arithmetic means of the increments. If all the speeds are equal (in particular if $n = 1$), the MLE does not exist and the function returns `NA` with a warning; this has probability zero when $n \geq 2$.

The root is computed by `stats::uniroot()` in $\ln a$. Kahle, Mercier and Paroissin (2016, remark 2.50) recommend a moment estimator as starting point. The mean and the variance of the gamma process are at/b and at/b^2 (Kahle et al., 2016, section 2.7), that is $\mu = a/b$ and $\nu = a/b^2$ per unit of time, so that $a = \mu^2/\nu$. The search starts from $a_{MM} = \hat{\mu}^2/\hat{\nu}$, where

$$\hat{\mu} = \frac{\sum_{j=1}^{n} \Delta x_j}{\sum_{j=1}^{n} \Delta t_j} = \frac{S}{T}, \qquad \hat{\nu} = \frac{1}{n}\sum_{j=1}^{n} \frac{(\Delta x_j - \hat{\mu}\Delta t_j)^2}{\Delta t_j}$$

are the estimators of the mean and of the variance per unit of time of Kahle et al. (2016, section 1.6.1, equation [1.19]), given there for the Wiener process. Since the root of g is unique, it does not depend on this starting point, which only makes the computation faster.

References

Kahle, W., Mercier, S. and Paroissin, C. (2016). *Degradation Processes in Reliability*. ISTE and Wiley, London. Definitions 2.1 and 2.24, section 1.6.1 (equation [1.19]) and section 2.7 (equations [2.42] to [2.45] and remark 2.50).

Alzer, H. (1997). On some inequalities for the gamma and psi functions. *Mathematics of Computation*, 66(217), 373-389. doi:10.1090/S0025571897008077. Equation (2.2) and Theorem 1.

van Noortwijk, J. M. (2009). A survey of the application of gamma processes in maintenance. *Reliability Engineering & System Safety*, 94(1), 2-21. doi:10.1016/j.ress.2007.03.019

See Also

`wiener_mle()`, `ig_mle()`

Examples

```
# 1. One trajectory inspected every time unit up to 50 (a = 2, b = 3)
t <- DegradSim::inspection_times("regular", n = 50, tmax = 50)$time
set.seed(1)
x <- DegradSim::sim_gamma(t, a = 2, b = 3)
gamma_mle(x)

# The same trajectory given as two vectors: degradation levels, then times
gamma_mle(x[, "X1"], t)

# 2. One trajectory with irregular inspections, more and more frequent
t <- DegradSim::inspection_times("decreasing", n = 50, tmax = 50)$time
x <- DegradSim::sim_gamma(t, a = 2, b = 3)
gamma_mle(x)
```

```
# 3. Ten trajectories with the same inspection times
t <- DegradSim::inspection_times("regular", n = 10, tmax = 50)$time
x <- DegradSim::sim_gamma(t, a = 2, b = 3, n_traj = 10)
gamma_mle(x)

# 4. Ten trajectories, each with its own inspection times
plan <- DegradSim::inspection_times("increasing", n = 5:14, n_traj = 10, tmax = 50)
x <- DegradSim::sim_gamma(plan, a = 2, b = 3)
gamma_mle(x)

# 5. Estimates with 95 % confidence intervals
gamma_mle(x, ci = TRUE)

# 6. Your own data: a list of data frames, one per trajectory,
#    with the times in the first column and the levels in the second
liste <- list(data.frame(t = c(0, 10, 20, 30), X = c(0, 6.1, 13.4, 19.8)),
              data.frame(t = c(0, 15, 30), X = c(0, 9.7, 20.5)))
gamma_mle(liste)

# The gamma process only increases: a decrease stops the estimation
try(gamma_mle(c(0, 2, 1.5, 3), c(0, 1, 2, 3)))
```

`ig_mle` *Maximum likelihood estimation for the inverse Gaussian process*

Description

Computes the maximum likelihood estimator (MLE) of the parameters μ and η of the inverse Gaussian (IG) process from one or several trajectories observed at regular or irregular inspection times. The MLE is obtained in closed form, without numerical optimization.

Usage

```
ig_mle(x, t = NULL, ci = FALSE, level = 0.95)
```

Arguments

`x`	Degradation levels of one trajectory, with strictly positive increments; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels. The increments can also be given directly, with the function `increments_data()`.
`t`	Inspection times of the trajectory, strictly increasing. Not used when `x` is a list.
`ci`	`FALSE` (default) for the point estimates only; `TRUE` to add asymptotic confidence intervals, computed as explained in the section Confidence intervals of levyfit-package.
`level`	Confidence level of the intervals, between 0 and 1 (default 0.95). Used only when `ci = TRUE`.

Value

A named vector with the maximum likelihood estimates `mu` $= \hat{\mu}$ and `eta` $= \hat{\eta}$, and the unbiased estimate `eta_unbiased` $= \tilde{\eta}$ of η.

With `ci = TRUE`, a data frame with one row per parameter and the columns `parameter`, `estimate`, `lower` and `upper` (bounds of the confidence interval). When the estimator does not exist, the estimates (and bounds) are `NA`, with a warning.

Model and data

$IG(m, \lambda)$ denotes the inverse Gaussian distribution with mean m and shape λ. The IG process is increasing, with independent increments $X_t - X_s \sim IG(\mu(t-s), \eta(t-s)^2)$, so that $E[X_t] = \mu t$ and $Var(X_t) = \mu^3 t/\eta$. The density of an increment over a duration Δt is

$$f_{\Delta t}(x; \mu, \eta) = \Delta t \sqrt{\frac{\eta}{2\pi x^3}} \exp\Big(- \frac{\eta(x - \mu\Delta t)^2}{2\mu^2 x}\Big), \qquad x > 0.$$

This is the inverse Gaussian process of Wang and Xu (2010, section 1 and equation (1)) with mean function $\Lambda(t) = \mu t$: they write the increments $IG(\Delta\Lambda, \eta_{WX}\Delta\Lambda^2)$, with $\Delta\Lambda = \Lambda(t) - \Lambda(s)$, so that the parameter η used here is their η_{WX} multiplied by μ^2: $\eta = \eta_{WX}\mu^2$. The data are the n increments $(\Delta t_j, \Delta x_j)$, $j = 1, \ldots, n$, with $\Delta x_j > 0$, pooled over the trajectories, with $T = \sum_{j=1}^n \Delta t_j$ and $S = \sum_{j=1}^n \Delta x_j$ (see the section Notation of levyfit-package).

Likelihood

The increments being independent (Wang and Xu, 2010, section 2, equation (3)), with $\delta = 1/\mu$, the log-likelihood is

$$\ell(\mu, \eta) = c_0 + \frac{n}{2}\ln\eta - \frac{\eta}{2}\sum_{j=1}^n \frac{(\delta\Delta x_j - \Delta t_j)^2}{\Delta x_j}, \qquad c_0 = \sum_{j=1}^n \Big[\ln\Delta t_j - \frac{1}{2}\ln(2\pi) - \frac{3}{2}\ln\Delta x_j\Big].$$

Maximum likelihood estimator

Expanding the square gives

$$\sum_{j=1}^n \frac{(\delta\Delta x_j - \Delta t_j)^2}{\Delta x_j} = \frac{(\delta S - T)^2}{S} + \hat{Q}, \qquad \hat{Q} = \sum_{j=1}^n \frac{\Delta t_j^2}{\Delta x_j} - \frac{T^2}{S} \geq 0,$$

with $\hat{Q} = 0$ if and only if the speeds $\Delta x_j/\Delta t_j$ are all equal. For every η, ℓ is therefore maximal at $\delta = T/S$, that is $\hat{\mu} = S/T$. It remains to maximize $\frac{n}{2}\ln\eta - \frac{\eta}{2}\hat{Q}$, which gives $\hat{\eta} = n/\hat{Q}$. The MLE exists and is unique if and only if $\hat{Q} > 0$, which has probability one as soon as $n \geq 2$. Otherwise the likelihood is unbounded and the function returns `NA` with a warning.

These are the estimators of Tweedie (1957, section 3, equations (13) and (14)), who considers independent observations $x_j \sim IG(\mu, \lambda_0 w_j)$ with known weights w_j. A change of scale multiplies the mean and the shape of an inverse Gaussian variable by the same factor (Tweedie, 1957, section 1, p. 364), so that the speeds $x_j = \Delta x_j/\Delta t_j$ follow $IG(\mu, \eta\Delta t_j)$: this is his setting with $w_j = \Delta t_j$ and $\lambda_0 = \eta$. His estimators are then $\hat{\mu} = \sum_{j=1}^n w_j x_j / \sum_{j=1}^n w_j = S/T$ and $1/\hat{\lambda}_0 = \frac{1}{n}\sum_{j=1}^n w_j(1/x_j - 1/\hat{\mu}) = \hat{Q}/n$.

Bias

With the same correspondence, Tweedie (1957, equation (15) and section 4, equations (21) and (22), p. 367-369) shows that, for $n \geq 2$,

$$\hat{\mu} \sim IG(\mu, \eta T), \qquad \eta\hat{Q} \sim \chi^2_{n-1},$$

and that $\hat{\mu}$ and $\hat{Q}$ are independent. Hence $\hat{\mu}$ is unbiased and, since $\hat{\eta} = n\eta/V$ with $V = \eta\hat{Q} \sim \chi^2_{n-1}$ and $E[1/V] = 1/(k-2)$ for $V \sim \chi^2_k$, $E[\hat{\eta}] = \frac{n}{n-3}\eta$ for $n > 3$: $\hat{\eta}$ is biased upwards. The function also returns its unbiased version

$$\tilde{\eta} = \frac{n-3}{\hat{Q}} = \frac{n-3}{n}\hat{\eta},$$

which requires at least four increments (`NA` otherwise). With equal durations Δt, the speeds are i.i.d. $IG(\mu, \eta\Delta t)$ and $\tilde{\eta}\Delta t$ is the uniformly minimum variance unbiased estimator $(n-3)/V$ of their shape parameter given by Seshadri (1999, section 2.1, Table 2.1, $n \geq 4$).

References

Tweedie, M. C. K. (1957). Statistical properties of inverse Gaussian distributions. I. *The Annals of Mathematical Statistics*, 28(2), 362-377. doi:10.1214/aoms/1177706964. Sections 1, 3 and 4, equations (13) to (15), (21) and (22).

Seshadri, V. (1999). *The Inverse Gaussian Distribution: Statistical Theory and Applications*. Lecture Notes in Statistics, 137. Springer, New York. doi:10.1007/9781461214564. Section 2.1, Table 2.1.

Wang, X. and Xu, D. (2010). An inverse Gaussian process model for degradation data. *Technometrics*, 52(2), 188-197. doi:10.1198/TECH.2009.08197. Sections 1 and 2, equations (1) and (3).

Ye, Z.-S. and Chen, N. (2014). The inverse Gaussian process as a degradation model. *Technometrics*, 56(3), 302-311. doi:10.1080/00401706.2013.830074

See Also

`wiener_mle()`, `gamma_mle()`

Examples

```
# 1. One trajectory inspected every time unit up to 50 (mu = 0.5, eta = 2)
t <- DegradSim::inspection_times("regular", n = 50, tmax = 50)$time
set.seed(1)
x <- DegradSim::sim_ig(t, mu = 0.5, eta = 2)
ig_mle(x)

# The same trajectory given as two vectors: degradation levels, then times
ig_mle(x[, "X1"], t)

# 2. One trajectory with irregular inspections, more and more frequent
t <- DegradSim::inspection_times("decreasing", n = 50, tmax = 50)$time
x <- DegradSim::sim_ig(t, mu = 0.5, eta = 2)
ig_mle(x)

# 3. Ten trajectories with the same inspection times
t <- DegradSim::inspection_times("regular", n = 10, tmax = 50)$time
x <- DegradSim::sim_ig(t, mu = 0.5, eta = 2, n_traj = 10)
ig_mle(x)
```

```
# 4. Ten trajectories, each with its own inspection times
plan <- DegradSim::inspection_times("increasing", n = 5:14, n_traj = 10, tmax = 50)
x <- DegradSim::sim_ig(plan, mu = 0.5, eta = 2)
ig_mle(x)

# 5. Estimates with 95 % confidence intervals
ig_mle(x, ci = TRUE)

# 6. Your own data: a list of data frames, one per trajectory,
#    with the times in the first column and the levels in the second
liste <- list(data.frame(t = c(0, 10, 20, 30), X = c(0, 5.2, 9.8, 15.3)),
              data.frame(t = c(0, 15, 30), X = c(0, 7.4, 14.6)))
ig_mle(liste)

# The inverse Gaussian process only increases: a decrease stops the estimation
try(ig_mle(c(0, 2, 1.5, 3), c(0, 1, 2, 3)))
```

`increments_data` *Data given by their increments*

Description

Builds the data of the estimation functions (`gamma_mle()`, `wiener_mle()`, `ig_mle()`, `vg_mle()`, `meixner_mle()`) from increments instead of degradation levels: the increments Δx_j over the durations Δt_j, $j = 1, \ldots, n$, of one or several trajectories. The likelihood of these processes depends on the data only through the increments. This is the fourth form of the data accepted by the estimation functions, after the three forms based on the degradation levels (see the section Data of levyfit-package).

Usage

```
increments_data(dx, dt = 1)
```

Arguments

`dx`	Numeric vector of increments.
`dt`	Durations of the increments: one positive number (all the increments over the same duration, 1 by default) or a vector of the same length as `dx`.

Details

Giving the increments directly keeps all their precision. In R, `.Machine$double.eps` (about 2.2×10^{-16}) is the smallest positive number x such that $1 + x \neq 1$ (see base::.Machine), so that an increment smaller than about 10^{-16} times the level can be rounded to 0 when computed as a difference of levels; this happens for instance with the increments reconstructed by the bridges of the package `levyBridge` on a fine grid.

Value

An object of class `"levy_increments"`, to be given as argument `x` of an estimation function (without `t`).

Examples

```
set.seed(1)
# Gamma increments over a time step 0.1, some of them very small
dx <- rgamma(200, shape = 2 * 0.1, rate = 3)
min(dx)
gamma_mle(increments_data(dx, dt = 0.1))

# Same estimate as from the levels, when no increment is lost
gamma_mle(c(0, cumsum(dx)), seq(0, 20, by = 0.1))

# Increments over different durations
dt <- runif(50, 0.5, 2)
wiener_mle(increments_data(rnorm(50, 0.5 * dt, 1.5 * sqrt(dt)), dt))
```

`meixner_mle` *Maximum likelihood estimation for the Meixner process*

Description

Computes the maximum likelihood estimator of the four parameters (a, b, d, m) of the Meixner process from one or several trajectories observed at regular or irregular inspection times. The likelihood is maximized numerically from several starting points.

Usage

```
meixner_mle(x, t = NULL, ci = FALSE, level = 0.95, n_start = 5)
```

Arguments

`x`	Degradation levels of one trajectory; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels. The increments can also be given directly, with the function `increments_data()`.
`t`	Inspection times of the trajectory, strictly increasing. Not used when `x` is a list.
`ci`	`FALSE` (default) for the point estimates only; `TRUE` to add asymptotic confidence intervals, computed as explained in the section Confidence intervals of levyfit-package.
`level`	Confidence level of the intervals, between 0 and 1 (default 0.95). Used only when `ci = TRUE`.
`n_start`	Number of starting points built from $\hat{\kappa}_1$ and $\hat{\kappa}_2$ (see Algorithm), in addition to the moment estimates.

Value

A named vector with the estimates `a`, `b`, `d` and `m`.

With `ci = TRUE`, a data frame with one row per parameter and the columns `parameter`, `estimate`, `lower` and `upper` (bounds of the confidence interval). When the estimator does not exist, the estimates (and bounds) are `NA`, with a warning.

Source and notation

The results on the Meixner distribution used below come from Grigoletto and Provasi (2009), who treat an i.i.d. sample of this distribution. The distribution written here $Meixner(a, b, d, m)$ is their $MXN(m, a, b, d)$: m is a location parameter, $a > 0$ a scale parameter, and $b \in (-\pi, \pi)$ and $d > 0$ are shape parameters (Grigoletto and Provasi, 2009, section 2); the same notation is used by Schoutens (2002), whose Meixner process is used here. The extension of the results of Grigoletto and Provasi (2009) to increments over unequal durations, and the algorithm, are choices of this implementation; they are marked as such.

Model and data

The density of $Meixner(a, b, d, m)$ is (Grigoletto and Provasi, 2009, section 2)

$$f(x) = \frac{(2\cos(b/2))^{2d}}{2a\pi\,\Gamma(2d)} \exp\Big(\frac{b(x-m)}{a}\Big)\Big|\Gamma\Big(d + i\,\frac{x-m}{a}\Big)\Big|^2, \qquad x \in (-\infty, \infty),$$

where i is the imaginary unit. Its characteristic function is $e^{imu}\big(\cos(b/2)/\cosh((au - ib)/2)\big)^{2d}$ (Schoutens, 2002, section 2), so that the distribution is infinitely divisible. Its mean and variance are $m + ad\tan(b/2)$ and $a^2 d/(1 + \cos b)$ (Grigoletto and Provasi, 2009, equation (1)).

The Meixner process X is the Levy process associated with this distribution: $X_0 = 0$, its increments are independent and stationary, and $X_s \sim Meixner(a, b, ds, ms)$ (Schoutens, 2002, section 2, where the increment over $[s, s + t]$ has the characteristic function of $Meixner(a, b, d, m)$ raised to the power t): an increment over a duration Δt has the density $f_{\Delta t}$ obtained by replacing d by $d\Delta t$ and m by $m\Delta t$ in f. The mean and the variance of the process per unit of time are $\kappa_1 = m + ad\tan(b/2)$ and $\kappa_2 = a^2 d/(1 + \cos b)$.

The data are the n increments $(\Delta t_j, \Delta x_j)$, $j = 1, \ldots, n$, pooled over the trajectories, with $T = \sum_{j=1}^n \Delta t_j$ and $S = \sum_{j=1}^n \Delta x_j$ (see the section Notation of levyfit-package). To shorten the formulas, let

$$s_j = d\Delta t_j, \qquad y_j = \frac{\Delta x_j - m\Delta t_j}{a}, \qquad j = 1, \ldots, n;$$

y_j is the standardized value $z_j = (x_j - m)/a$ of Grigoletto and Provasi (2009) (section 4.2) for an increment over Δt_j.

Likelihood and score

The increments being independent, the log-likelihood is

$$\ell(a, b, d, m) = \sum_{j=1}^n \Big[2s_j \ln(2\cos(b/2)) - \ln(2a\pi) - \ln\Gamma(2s_j) + by_j + \ln|\Gamma(s_j + iy_j)|^2\Big].$$

With equal durations $\Delta t_j = 1$, it is n times the average log-likelihood $\ell_n(\theta)$ of Grigoletto and Provasi (2009, equation (6)); with unequal durations, it is its extension to the process.

The first derivatives (the score) are explicit (Grigoletto and Provasi, 2009, appendix A). Grigoletto and Provasi (2009) write them with $\Psi_j^+ = \psi(d+iz_j)+\psi(d-iz_j)$ and $\Psi_j^- = \psi(d+iz_j)-\psi(d-iz_j)$, ψ being the digamma function. Here, with $\psi_j = \psi(s_j + iy_j)$ and its real and imaginary parts $\mathrm{Re}\,\psi_j$ and $\mathrm{Im}\,\psi_j$, these are $\Psi_j^+ = 2\,\mathrm{Re}\,\psi_j$ and $i\,\Psi_j^- = -2\,\mathrm{Im}\,\psi_j$, and the score is

$$\frac{\partial \ell}{\partial a} = \frac{1}{a}\sum_{j=1}^n \big(-1 - by_j + 2y_j\,\mathrm{Im}\,\psi_j\big),$$

$$\frac{\partial \ell}{\partial b} = \sum_{j=1}^n y_j - \tan(b/2)\sum_{j=1}^n s_j,$$

$$\frac{\partial \ell}{\partial d} = \sum_{j=1}^{n} \Delta t_j \big(2\ln(2\cos(b/2)) - 2\psi(2s_j) + 2\operatorname{Re}\psi_j\big),$$

$$\frac{\partial \ell}{\partial m} = \frac{1}{a}\sum_{j=1}^{n} \Delta t_j \big(2\operatorname{Im}\psi_j - b\big).$$

Profile in b

For fixed (a, d, m), $\partial\ell/\partial b = 0$ has the unique solution

$$\hat{b}(a, d, m) = 2\arctan\frac{S - mT}{adT},$$

which is the lemma of Grigoletto and Provasi (2009, section 4.2), $\hat{b} = 2\tan^{-1}(\bar{z}/\hat{d})$, written for increments over unequal durations. It is a maximum, since $\partial^2\ell/\partial b^2 = -\sum_{j=1}^{n} s_j/(1+\cos b) < 0$ (Grigoletto and Provasi, 2009, appendix A). It makes the expected total increment $\kappa_1 T$ equal to S. The maximization is therefore done in dimension 3, on the profile log-likelihood $\ell_p(a, d, m) = \ell(a, \hat{b}(a, d, m), d, m)$, as suggested by Grigoletto and Provasi (2009, section 4.2).

Method of moments

With unequal durations (extension of Grigoletto and Provasi, 2009, section 4.1), the mean and the higher cumulants of the process per unit of time are estimated by $k_1 = S/T$ and, with the residuals $e_j = \Delta x_j - k_1 \Delta t_j$,

$$k_2 = \frac{1}{T}\sum_{j=1}^{n} e_j^2, \qquad k_3 = \frac{1}{T}\sum_{j=1}^{n} e_j^3, \qquad k_4 = \frac{1}{T}\Big(\sum_{j=1}^{n} e_j^4 - 3k_2^2 \sum_{j=1}^{n} \Delta t_j^2\Big).$$

Let $\gamma_1 = k_3/k_2^{3/2}$ and $\gamma_2 = k_4/k_2^2$ be the skewness and the excess kurtosis of X_1. Since the skewness of $Meixner(a, b, d, m)$ is $\sin b/\sqrt{d(1+\cos b)}$ and its kurtosis $3 + (2-\cos b)/d$ (Grigoletto and Provasi, 2009, equation (1)), that is $\gamma_1^2 = (1-\cos b)/d$ and $\gamma_2 = (2-\cos b)/d$, the moment estimates are (Grigoletto and Provasi, 2009, section 4.1, with the excess kurtosis)

$$d = \frac{1}{\gamma_2 - \gamma_1^2}, \quad b = \operatorname{sign}(\gamma_1)\arccos(2 - d\gamma_2), \quad a = \sqrt{k_2(1+\cos b)/d}, \quad m = k_1 - ad\tan(b/2).$$

They exist if and only if $d > 0$ and $-1 < 2 - d\gamma_2 \le 1$, so that $b \in (-\pi, \pi)$. Since $2 - d\gamma_2 = (\gamma_2 - 2\gamma_1^2)/(\gamma_2 - \gamma_1^2)$, this is $\gamma_2 > \frac{3}{2}\gamma_1^2$. When b ranges over $(-\pi, \pi)$, the ratio $\gamma_2/\gamma_1^2 = (2 - \cos b)/(1 - \cos b)$ takes all the values larger than $3/2$: the condition says that the sample skewness and kurtosis are those of a Meixner distribution. Grigoletto and Provasi (2009, section 4.1) write that the estimates do not exist when the kurtosis $\gamma_2 + 3$ is smaller than $2\gamma_1^2 + 3$, that is $\gamma_2 < 2\gamma_1^2$. The bound 2 instead of $3/2$ corresponds to $|b| = \pi/2$: it excludes the Meixner distributions with $\pi/2 < |b| < \pi$, for which the estimates exist. With equal durations Δt, they are the estimates of Grigoletto and Provasi (2009, section 4.1) computed on the increments, with the variance divided by n, and with d and m divided by Δt.

Algorithm

Grigoletto and Provasi (2009, section 4.2) propose to maximize the likelihood by a Newton-type algorithm with the first and second derivatives, and note that the moment estimates can be used as starting points. These estimates do not always exist (see Method of moments). Over a long time, the Meixner process is close to a Brownian motion (Kawai and Masuda, 2011, section 1): with long durations Δt_j, the increments are close to normal, their skewness and excess kurtosis are close to 0,

and the condition of existence may fail. This implementation therefore uses several starting points, and the first derivatives only. The additional starting points, the bounds and the thresholds below are choices of this implementation, without a source.

1. To build the additional starting points of step 2, estimate the mean κ_1 and the variance κ_2 of the process per unit of time (see Model and data) by

$$\hat\kappa_1 = \frac{\sum_{j=1}^n \Delta x_j}{\sum_{j=1}^n \Delta t_j} = \frac{S}{T}, \qquad \hat\kappa_2 = \frac{1}{n}\sum_{j=1}^n \frac{(\Delta x_j - \hat\kappa_1 \Delta t_j)^2}{\Delta t_j},$$

   and compute the mean duration $\overline{\Delta t} = T/n$. For the Wiener process, $\hat\kappa_1$ and $\hat\kappa_2$ are the maximum likelihood estimators of the drift and of the variance of Kahle, Mercier and Paroissin (2016, section 1.6.1, equation [1.19]). They only use the mean and the variance of the increments, which are $E[\Delta x_j] = \kappa_1 \Delta t_j$ and $Var(\Delta x_j) = \kappa_2 \Delta t_j$ for the Meixner process too (see Model and data); a direct computation then gives $E[\hat\kappa_1] = \kappa_1$ and $E[\hat\kappa_2] = (1 - 1/n)\kappa_2$.

2. Build the starting points (a_0, d_0, m_0) (the value of b is always $\hat b(a, d, m)$, see Profile in b):
   - the moment estimates of a, d, m (see Method of moments), when they exist, as suggested by Grigoletto and Provasi (2009, section 4.2);
   - the point $b_0 = 0$, $d_0 = 1/\overline{\Delta t}$;
   - `n_start` - 1 random points: b_0 drawn uniformly on $(-\pi + 0.1, \pi - 0.1)$, and $d_0 = u/\overline{\Delta t}$ with $\ln u$ drawn uniformly on $(\ln 0.2, \ln 5)$.

   For the last two kinds of points, a_0 and m_0 are then chosen so that the mean and the variance of the model are equal to their estimates, that is $\kappa_1 = \hat\kappa_1$ and $\kappa_2 = \hat\kappa_2$:

$$a_0 = \sqrt{\frac{\hat\kappa_2(1 + \cos b_0)}{d_0}}, \qquad m_0 = \hat\kappa_1 - a_0 d_0 \tan(b_0/2).$$

3. From each starting point, maximize ℓ_p by L-BFGS-B (`stats::optim()`) on $(\ln a, \ln d, m)$, with the exact gradient, and with the bounds $10^{-3}\sqrt{\hat\kappa_2} \le a \le 10^3\sqrt{\hat\kappa_2}$, $10^{-3}/\max_j \Delta t_j \le d \le 10^3/\min_j \Delta t_j$ and $|m - \hat\kappa_1| \le 100\sqrt{\hat\kappa_2}$.

4. Discard the solutions at the edge of the parameter space: those with $|\hat b| > \pi - 0.1$, with $d \max_j \Delta t_j > 1000$, or on one of the bounds of step 3. Among the remaining solutions, keep the one of largest likelihood (when several reach the same maximum, one where L-BFGS-B converged). When all the solutions are discarded, the function returns `NA` with a warning. A warning is also given when L-BFGS-B did not converge at the solution kept, or when a discarded solution has a larger likelihood than the solution kept.

Because of the random starting points, call `set.seed()` before `meixner_mle()` to reproduce a result exactly.

Grigoletto and Provasi (2009, section 4.2) find that the convergence of the estimator to its asymptotic normal distribution is very slow, with a speed inversely related to d, and that inference based on asymptotic normality must be carefully evaluated even for medium samples.

Hessian

The second derivatives of ℓ are explicit (Grigoletto and Provasi, 2009, appendix A, where they are written with $\Upsilon_j^{\pm} = \psi'(d + iz_j) \pm \psi'(d - iz_j)$, ψ' being the trigamma function). Here, with $\psi'_j = \psi'(s_j + iy_j)$,

$$G_j = b - 2\,\mathrm{Im}\,\psi_j, \qquad R_j = \mathrm{Re}\,\psi'_j, \qquad I_j = \mathrm{Im}\,\psi'_j, \qquad j = 1, \dots, n,$$

so that $\Upsilon_j^+ = 2R_j$ and $\Upsilon_j^- = 2iI_j$, they are

$$\frac{\partial^2\ell}{\partial a^2} = \frac{1}{a^2}\sum_{j=1}^{n}\big(1 - 2R_j y_j^2 + 2G_j y_j\big), \qquad \frac{\partial^2\ell}{\partial a\partial b} = -\frac{1}{a}\sum_{j=1}^{n} y_j, \qquad \frac{\partial^2\ell}{\partial a\partial d} = \frac{2}{a}\sum_{j=1}^{n} I_j y_j \Delta t_j,$$

$$\frac{\partial^2\ell}{\partial a\partial m} = \frac{1}{a^2}\sum_{j=1}^{n}\Delta t_j\big(G_j - 2R_j y_j\big), \qquad \frac{\partial^2\ell}{\partial b^2} = -\frac{1}{1+\cos b}\sum_{j=1}^{n} s_j,$$

$$\frac{\partial^2\ell}{\partial b\partial d} = -T\tan(b/2), \qquad \frac{\partial^2\ell}{\partial b\partial m} = -\frac{T}{a}, \qquad \frac{\partial^2\ell}{\partial d^2} = \sum_{j=1}^{n}\Delta t_j^2\big(2R_j - 4\psi'(2s_j)\big),$$

$$\frac{\partial^2\ell}{\partial d\partial m} = \frac{2}{a}\sum_{j=1}^{n} I_j\Delta t_j^2, \qquad \frac{\partial^2\ell}{\partial m^2} = -\frac{2}{a^2}\sum_{j=1}^{n} R_j\Delta t_j^2.$$

With `ci = TRUE`, the observed Fisher information is minus this Hessian at the estimate, as proposed by Grigoletto and Provasi (2009, section 4.2), instead of the numerical Hessian used by the other models.

References

Grigoletto, M. and Provasi, C. (2009). Simulation and estimation of the Meixner distribution. *Communications in Statistics - Simulation and Computation*, 38(1), 58-77. doi:10.1080/03610910802395679

Kahle, W., Mercier, S. and Paroissin, C. (2016). *Degradation Processes in Reliability*. ISTE and Wiley, London. Section 1.6.1 (equation [1.19]).

Kawai, R. and Masuda, H. (2011). On the local asymptotic behavior of the likelihood function for Meixner Levy processes under high-frequency sampling. *Statistics and Probability Letters*, 81(4), 460-469.

Schoutens, W. (2002). The Meixner process: theory and applications in finance. EURANDOM report 2002-004. Section 2.

See Also

`vg_mle()`

Examples

```
# 1. One trajectory with 5000 inspections up to 500 (step 0.1)
#    (a = 1, b = 0.3, d = 1.5, m = 0.2)
t <- DegradSim::inspection_times("regular", n = 5000, tmax = 500)$time
set.seed(1)
x <- DegradSim::sim_meixner(t, a = 1, b = 0.3, d = 1.5, m = 0.2)
meixner_mle(x)

# The same trajectory given as two vectors: degradation levels, then times
meixner_mle(x[, "X1"], t)

# 2. The same duration with a step 1 (500 increments); comparison of the
#    mean and the variance of an increment over one time unit, true and
#    fitted
t1 <- DegradSim::inspection_times("regular", n = 500, tmax = 500)$time
set.seed(1)
```

```
x1 <- DegradSim::sim_meixner(t1, a = 1, b = 0.3, d = 1.5, m = 0.2)
p <- meixner_mle(x1)
p
c(true = 0.2 + 1.5 * tan(0.15), fitted = p[["m"]] + p[["a"]] * p[["d"]] * tan(p[["b"]] / 2))
c(true = 1.5 / (2 * cos(0.15)^2), fitted = p[["a"]]^2 * p[["d"]] / (2 * cos(p[["b"]] / 2)^2))

# 3. Twenty trajectories with the same inspection times (step 0.1)
t <- DegradSim::inspection_times("regular", n = 200, tmax = 20)$time
x <- DegradSim::sim_meixner(t, a = 1, b = 0.3, d = 1.5, m = 0.2, n_traj = 20)
meixner_mle(x)

# 4. Estimates with 95 % confidence intervals
meixner_mle(x, ci = TRUE)

# 5. Ten increments over long intervals: when no acceptable maximum is
#    found, the function gives a warning and NA estimates
t10 <- DegradSim::inspection_times("regular", n = 10, tmax = 100)$time
y <- DegradSim::sim_meixner(t10, a = 1, b = 0.3, d = 1.5, m = 0.2)
meixner_mle(y)
```

`vg_mle` *Maximum likelihood estimation for the variance-gamma process*

Description

Computes the maximum likelihood estimator (MLE) of the parameters (θ, σ, ν) of the variance-gamma (VG) process from one or several trajectories observed at regular or irregular inspection times. There is no closed form: the MLE is computed by an EM algorithm, whose steps are explicit or reduce to an equation with a unique root, followed by a quasi-Newton refinement.

Usage

```
vg_mle(x, t = NULL, ci = FALSE, level = 0.95)
```

Arguments

`x`	Degradation levels of one trajectory, with non-zero increments; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels. The increments can also be given directly, with the function `increments_data()`.
`t`	Inspection times of the trajectory, strictly increasing. Not used when `x` is a list.
`ci`	`FALSE` (default) for the point estimates only; `TRUE` to add asymptotic confidence intervals, computed as explained in the section Confidence intervals of levyfit-package.
`level`	Confidence level of the intervals, between 0 and 1 (default 0.95). Used only when `ci = TRUE`.

Value

A named vector with the maximum likelihood estimates `theta`, `sigma` and `nu`.

With `ci = TRUE`, a data frame with one row per parameter and the columns `parameter`, `estimate`, `lower` and `upper` (bounds of the confidence interval). When the estimator does not exist, the estimates (and bounds) are `NA`, with a warning.

Model and data

Let γ be a gamma process such that γ_s has mean s and variance νs (shape s/ν, rate $1/\nu$), and W an independent standard Brownian motion. The VG process is

$$X_s = \theta\gamma_s + \sigma W_{\gamma_s} :$$

a Brownian motion with drift θ and diffusion σ, observed at the random time γ_s instead of the time s (Madan and Seneta, 1990, section III.A, equation (7), for $\theta = 0$; Seneta, 2004, equations (1) and (2); Bee, Dickson and Santi, 2018, section 2, where the increments of the random time over one unit of time follow a gamma distribution with shape and rate both equal to their ν, that is $1/\nu$ here). It is a Levy process that can decrease.

Given the random time, an increment over a duration Δt is $\theta G + \sigma\sqrt{G}Z$, with $G \sim \Gamma(\Delta t/\nu, 1/\nu)$ and $Z \sim \mathcal{N}(0,1)$ independent. This is the representation (2.18) of Fischer, Gaunt and Sarantsev (2023) with $S = 2G/\nu$, so that the increment follows their distribution $VG(r, \theta_F, \sigma_F, 0)$ with $r = 2\Delta t/\nu$, $\theta_F = \theta\nu/2$ and $\sigma_F = \sigma\sqrt{\nu/2}$. Their density (1.1) then gives, with $\lambda = \Delta t/\nu$, $c = \sqrt{\theta^2 + 2\sigma^2/\nu}$ and $\kappa = c/\sigma^2$,

$$f_{\Delta t}(x;\theta,\sigma,\nu) = \frac{2e^{\theta x/\sigma^2}}{\sqrt{2\pi}\,\sigma\,\nu^{\lambda}\Gamma(\lambda)}\left(\frac{|x|}{c}\right)^{\lambda-1/2} K_{|\lambda-1/2|}(\kappa|x|), \qquad x \neq 0,$$

where K_p is the modified Bessel function of the second kind of order p, defined by $K_p(x) = \int_0^\infty e^{-x\cosh t}\cosh(pt)\,dt$ (Fischer et al., 2023, appendix A), so that $K_{-p} = K_p$. Some authors call K_p the modified Bessel function of the third kind; Seneta (2004, section 3) points out this ambiguity of terminology. For $\Delta t = 1$, this is the density (15) of Seneta (2004) with his location $c = 0$, and the density (1) of Bee et al. (2018) with their location $\mu = 0$, their γ equal to θ and their ν equal to $1/\nu$.

The data are the n increments $(\Delta t_j, \Delta x_j)$, $j = 1, \ldots, n$, with $\Delta x_j \neq 0$, pooled over the trajectories, with $T = \sum_{j=1}^n \Delta t_j$ and $S = \sum_{j=1}^n \Delta x_j$ (see the section Notation of levyfit-package).

Likelihood

The increments being independent, the log-likelihood is

$$\ell(\theta,\sigma,\nu) = \sum_{j=1}^{n} \ln f_{\Delta t_j}(\Delta x_j;\theta,\sigma,\nu),$$

which is equation (4) of Bee et al. (2018) for $\Delta t_j = 1$.

Why an EM algorithm. The density contains the modified Bessel function K_p, whose order depends on ν, and the likelihood cannot be maximized in closed form. Direct maximization is likely to be computationally expensive (Madan and Seneta, 1990, section II.F), the log-likelihood can have many local optima (Belhaj Salem, Fouladirad and Deloux, 2022, section 2.2), and Seneta (2004, section 3) reports, for the model with a location parameter, a likelihood surface that appeared flat near the moment estimates. Standard optimization algorithms applied directly to this likelihood, such as the Nelder-Mead algorithm of `stats::optim()`, are very sensitive to the starting values and give highly unstable estimates (Bee et al., 2018, section 1, after Cervellera and Tucci). On the other hand, the maximization would be easy if the random times were known: this is why the MLE is computed by an EM algorithm, which treats the random times as missing data (Bee et al., 2018, sections 1 and 3.2).

When the estimator does not exist

The VG model has two other models as limits, which are not in its parameter space ($\nu > 0$, $\sigma > 0$). The likelihood can increase towards one of these limits, without a maximum inside the parameter

space: the MLE then does not exist, and the function returns NA with a warning that names the model to use instead.

- $\nu \to 0$: with the parameters above, the characteristic function of an increment (Fischer et al., 2023, equation (2.12)) is $\varphi(u) = (1 - i\theta\nu u + \sigma^2\nu u^2/2)^{-\Delta t/\nu}$, which tends to $\exp(\Delta t(i\theta u - \sigma^2 u^2/2))$, the characteristic function of $\mathcal{N}(\theta\Delta t, \sigma^2\Delta t)$. By Levy's continuity theorem (van der Vaart, 1998, Theorem 2.13), each increment converges in distribution to this normal distribution: the VG process tends to the Wiener process with drift θ and diffusion σ. For $\theta = 0$, Seneta (2004, section 3) gives the same limit. The function concludes that the likelihood increases towards this limit when ν falls below $\max_j \Delta t_j/300$ during the EM algorithm, or when L-BFGS-B stops on this bound. The value 300 is a choice of this implementation, without a source. Its meaning: for $\theta = 0$, the kurtosis of an increment over a duration Δt is $3(1 + \nu/\Delta t)$ (Madan and Seneta, 1990, section III.C), so that at the threshold the excess kurtosis of the increment over the longest duration is $3/300 = 0.01$: this increment is then almost normal. Use `wiener_mle()`.
- $\sigma \to 0$: when $\theta > 0$, an increment tends in distribution to the gamma distribution with shape $\Delta t/\nu$ and rate $1/(\theta\nu)$ (Fischer et al., 2023, section 2.5.1, item 2, with their r equal to $\Delta t/\nu$ and their λ equal to $1/(\theta\nu)$): the VG process tends to the gamma process with shape $1/\nu$ and rate $1/(\theta\nu)$. The density converges too: applying the limiting form (A.65) of K_p to the density (1.1) of Fischer et al. (2023) shows that, as $\sigma \to 0$, the VG density tends to this gamma density at every $x > 0$, and to 0 at every $x < 0$. The VG likelihood therefore tends to the gamma likelihood. When all the increments are positive, the function compares the VG likelihood at the estimate with the maximum of the gamma likelihood; if the VG likelihood is not larger, the function concludes that the likelihood increases towards this limit. Use `gamma_mle()`.

When a computation of the EM algorithm gives a value that is not finite, the function returns NA with a warning that reports this numerical failure, without naming a limit.

EM algorithm

The EM algorithm in general (Bee et al., 2018, section 3.2). The data are completed by unobserved variables (the missing data), chosen so that the log-likelihood of the complete data is simple. Starting from initial values of the parameters, the algorithm repeats two steps:

- E step (expectation): compute the expected value of the complete log-likelihood given the observed data, at the current value of the parameters;
- M step (maximization): maximize this expected value with respect to the parameters, which gives their new value.

The likelihood of the observed data increases at each iteration (Bee et al., 2018, section 3.2).

Application to the VG process. The algorithm used here is the one of Bee et al. (2018, section 3.2), who follow Nitithumbundit and Chan (2015), written for increments over unequal durations and without location parameter. The correspondence with the general algorithm is the following.

- Observed data: the increments Δx_j, $j = 1, \ldots, n$.
- Missing data: the increments G_j of the random time γ on the j-th interval.
- Complete log-likelihood: since

$$\Delta x_j \mid G_j \sim \mathcal{N}(\theta G_j, \sigma^2 G_j), \qquad G_j \sim \Gamma(\Delta t_j/\nu,\ 1/\nu),$$

  it is the sum of normal and gamma log-densities, and it only involves G_j, $1/G_j$ and $\ln G_j$ (Bee et al., 2018, equation (5)).
- E step: compute the expectations e_j^+, e_j^- and l_j of G_j, $1/G_j$ and $\ln G_j$ given Δx_j (see below).

- M step: update (θ, σ^2), then ν (see Steps below).

Distribution of the hidden times. Given Δx_j, G_j follows a generalized inverse Gaussian distribution $GIG(p_j, \chi_j, \omega)$ (Bee et al., 2018, section 3.2, after Embrechts, 1983), with density proportional to $g^{p_j-1}\exp(-(\chi_j/g + \omega g)/2)$, $g > 0$, where

$$p_j = \frac{\Delta t_j}{\nu} - \frac{1}{2}, \qquad \chi_j = \frac{\Delta x_j^2}{\sigma^2}, \qquad \omega = \frac{\theta^2}{\sigma^2} + \frac{2}{\nu}.$$

With $z_j = \sqrt{\chi_j \omega}$ and $\rho_j = \sqrt{\chi_j/\omega}$, its moments are ratios of Bessel functions (Bee et al., 2018, equations (6) to (8)):

$$e_j^+ = E[G_j] = \rho_j \frac{K_{p_j+1}(z_j)}{K_{p_j}(z_j)}, \qquad e_j^- = E[1/G_j] = \frac{1}{\rho_j}\frac{K_{p_j-1}(z_j)}{K_{p_j}(z_j)},$$

$$l_j = E[\ln G_j] = \ln \rho_j + \frac{\partial}{\partial p} \ln K_p(z_j)\Big|_{p=p_j},$$

the last derivative being computed numerically, as in Bee et al. (2018).

Steps.

1. E step: compute e_j^+, e_j^- and l_j, $j = 1, \ldots, n$, at the current parameters.
2. M step for (θ, σ^2), explicit (Bee et al., 2018, equations (10) and (11) with their location $\mu = 0$): with $E^+ = \sum_{j=1}^n e_j^+$,

$$\theta = \frac{S}{E^+}, \qquad \sigma^2 = \frac{1}{n}\Big(\sum_{j=1}^n \Delta x_j^2 e_j^- - \frac{S^2}{E^+}\Big).$$

3. M step for ν: with $a = 1/\nu$ and ψ the digamma function, the derivative of the complete log-likelihood in a is

$$q(a) = \sum_{j=1}^n \big[\Delta t_j l_j - e_j^+ + \Delta t_j \ln a + \Delta t_j - \Delta t_j\, \psi(a \Delta t_j)\big],$$

   which is the derivative given in the CM-step 2a of Bee et al. (2018) when $\Delta t_j = 1$. Since $\psi'(x) > 1/x$ (Alzer, 1997, Theorem 1), q is strictly decreasing; by the inequality (2.2) of Alzer (1997) it tends to $+\infty$ as $a \to 0$ and, by Jensen's inequality ($l_j < \ln e_j^+$) and $\ln u \le u - 1$, to a negative limit as $a \to \infty$: its unique root gives $\nu = 1/a$.
4. Repeat steps 1 to 3 until the increase of ℓ becomes small.

Start and end. As recommended by Seneta (2004, section 3), the algorithm starts from moment estimates.

- θ and σ: the Wiener MLE (see `wiener_mle()`), $\theta_0 = S/T$ and $\sigma_0^2 = \frac{1}{n}\sum_{j=1}^n (\Delta x_j - \theta_0 \Delta t_j)^2/\Delta t_j$, which estimate the mean and the variance of the process per unit of time. Seneta (2004, equations (23) and (24)) uses the same estimates for θ and σ^2 when the terms in θ^2, θ^3, θ^4 are ignored (his location c being 0 here).
- ν: the kurtosis of X_1 is $3(1+\nu)$ when these terms are ignored (Seneta, 2004, equation (26); exactly when $\theta = 0$, Madan and Seneta, 1990, sections II.B and II.E), so that $\nu_0 = \hat\gamma_2/3$, where $\hat\gamma_2 = k_4/k_2^2$ is the excess kurtosis of X_1 estimated with the residuals $e_j = \Delta x_j - \theta_0 \Delta t_j$ by $k_2 = \frac{1}{T}\sum_{j=1}^n e_j^2$ and $k_4 = \frac{1}{T}\big(\sum_{j=1}^n e_j^4 - 3k_2^2 \sum_{j=1}^n \Delta t_j^2\big)$. These estimators of the cumulants per unit of time, which take the unequal durations into account, are an extension made in this implementation: they use the fact that the cumulants of an increment are proportional

to its duration Δt, the logarithm of its characteristic function (see When the estimator does not exist) being proportional to Δt. When ν_0 is not larger than the threshold $\max_j \Delta t_j/300$ (see When the estimator does not exist), in particular when $\hat{\gamma}_2 \leq 0$, the moment estimate is not usable and the algorithm starts from $\nu_0 = T/n$, a choice of this implementation, without a source.

The maximization is then finished by L-BFGS-B on $(\theta, \ln\sigma, \ln\nu)$, started at the last EM iterate. If this refinement fails or does not increase the likelihood, the EM iterate is returned.

Numerical detail. The Bessel functions $K_p(z)$ are computed on the log scale; when `besselK()` overflows (large order p, that is small ν, and small z), $\ln K_p(z)$ is computed exactly by the recurrence on the order $K_{p+1}(z) = K_{p-1}(z) + (2p/z)K_p(z)$ (Olver et al., 2010, equation 10.29.1, with $\mathcal{Z}_\nu = e^{\nu\pi i}K_\nu$ as defined in section 10.25(ii)).

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See Also

`wiener_mle()`, `meixner_mle()`

Examples

```
# 1. One trajectory inspected every 2 time units up to 400
#    (theta = 0.5, sigma = 1.5, nu = 2)
t <- DegradSim::inspection_times("regular", n = 200, tmax = 400)$time
set.seed(1)
```

```
x <- DegradSim::sim_vg(t, theta = 0.5, sigma = 1.5, nu = 2)
vg_mle(x)

# The same trajectory given as two vectors: degradation levels, then times
vg_mle(x[, "X1"], t)

# 2. Twenty trajectories with the same inspection times
t <- DegradSim::inspection_times("regular", n = 20, tmax = 40)$time
x <- DegradSim::sim_vg(t, theta = 0.5, sigma = 1.5, nu = 2, n_traj = 20)
vg_mle(x)

# 3. Twenty trajectories, each with its own inspection times
plan <- DegradSim::inspection_times("inc-dec", n = 15:34, n_traj = 20, tmax = 40)
x <- DegradSim::sim_vg(plan, theta = 0.5, sigma = 1.5, nu = 2)
vg_mle(x)

# 4. Estimates with 95 % confidence intervals
vg_mle(x, ci = TRUE)

# 5. Increasing data, here from a gamma process: the likelihood increases
#    towards the gamma limit (sigma = 0): warning suggesting gamma_mle(), NA estimates
t100 <- DegradSim::inspection_times("regular", n = 100, tmax = 100)$time
y <- DegradSim::sim_gamma(t100, a = 2, b = 3)
vg_mle(y)
gamma_mle(y)
```

`wiener_mle` *Maximum likelihood estimation for the Wiener process*

Description

Computes the maximum likelihood estimator (MLE) of the drift μ and the variance σ^2 of the Wiener process $X_t = \mu t + \sigma B_t$, where B is a standard Brownian motion, from one or several trajectories observed at regular or irregular inspection times. The MLE is obtained in closed form, without numerical optimization.

Usage

```
wiener_mle(x, t = NULL, ci = FALSE, level = 0.95)
```

Arguments

`x`	Degradation levels of one trajectory; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels. The increments can also be given directly, with the function `increments_data()`.
`t`	Inspection times of the trajectory, strictly increasing. Not used when `x` is a list.
`ci`	`FALSE` (default) for the point estimates only; `TRUE` to add asymptotic confidence intervals, computed as explained in the section Confidence intervals of levyfit-package.
`level`	Confidence level of the intervals, between 0 and 1 (default 0.95). Used only when `ci = TRUE`.

Value

A named vector with the maximum likelihood estimates `mu` $= \hat{\mu}$ and `sigma2` $= \hat{\sigma}^2$, and the unbiased estimate `s2` $= s^2$ of σ^2.

With `ci = TRUE`, a data frame with one row per parameter and the columns `parameter`, `estimate`, `lower` and `upper` (bounds of the confidence interval). When the estimator does not exist, the estimates (and bounds) are `NA`, with a warning.

Model and data

The increments are independent and $X_t - X_s \sim \mathcal{N}(\mu(t-s), \sigma^2(t-s))$ for $s < t$ (Kahle, Mercier and Paroissin, 2016, definition 1.11 and equation [1.3]); they can be negative. The density of an increment over a duration Δt is

$$f_{\Delta t}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2\Delta t}} \exp\Big(- \frac{(x - \mu\Delta t)^2}{2\sigma^2 \Delta t}\Big), \qquad x \in (-\infty, \infty).$$

The data are the n increments $(\Delta t_j, \Delta x_j)$, $j = 1, \dots, n$, pooled over the trajectories, with $T = \sum_{j=1}^n \Delta t_j$ and $S = \sum_{j=1}^n \Delta x_j$ (see the section Notation of levyfit-package). The speeds are $r_j = \Delta x_j / \Delta t_j$.

Likelihood

The increments being independent, the log-likelihood is (Kahle et al., 2016, equation [1.14], with $x_0 = 0$ and $t_0 = 0$)

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{1}{2}\sum_{j=1}^n \ln \Delta t_j - \frac{n}{2}\ln\sigma^2 - \frac{Q(\mu)}{2\sigma^2}, \qquad Q(\mu) = \sum_{j=1}^n \frac{(\Delta x_j - \mu\Delta t_j)^2}{\Delta t_j}.$$

Maximum likelihood estimator

Completing the square gives $Q(\mu) = T(\mu - \hat{\mu})^2 + \hat{Q}$, with

$$\hat{\mu} = \frac{S}{T}, \qquad \hat{Q} = Q(\hat{\mu}) = \sum_{j=1}^n \Delta t_j (r_j - \hat{\mu})^2.$$

For every σ^2, ℓ is therefore maximal in μ at $\hat{\mu}$, which is the mean of the speeds weighted by $\Delta t_j / T$. It remains to maximize $-\frac{n}{2}\ln\sigma^2 - \hat{Q}/(2\sigma^2)$, which gives $\hat{\sigma}^2 = \hat{Q}/n$. These are the estimators of Kahle et al. (2016, section 1.6.1, equation [1.19]). The MLE exists and is unique if and only if $\hat{Q} > 0$, that is if the speeds r_j are not all equal, which has probability one as soon as $n \geq 2$. Otherwise (in particular if $n = 1$) the likelihood is unbounded and the function returns `NA` with a warning.

Equation [1.19] also writes $\hat{\mu}$ as the sum of the last observed levels divided by the sum of the last inspection times: with one trajectory, $\hat{\mu}$ only depends on the first and last observations, and the intermediate inspections only bring information on σ^2.

Bias

Whitmore (1995, section 3, equations (3.8a), (3.8b), (3.10a) and (3.10b)) considers increments $\Delta Y \sim \mathcal{N}_n(\delta\Delta t, \sigma^2\Psi)$ with a known matrix Ψ; his estimators are

$$\hat{\delta} = \frac{\Delta t' \Psi^{-1} \Delta Y}{\Delta t' \Psi^{-1} \Delta t}, \qquad \hat{\sigma}^2 = \frac{(\Delta Y - \hat{\delta}\Delta t)' \Psi^{-1} (\Delta Y - \hat{\delta}\Delta t)}{n},$$

and they are independent, with $\hat{\delta} \sim \mathcal{N}(\delta, \sigma^2/(\Delta t' \Psi^{-1} \Delta t))$ and $n\hat{\sigma}^2/\sigma^2 \sim \chi^2_{n-1}$. Here the increments are independent with variances $\sigma^2 \Delta t_j$, that is $\Psi = \mathrm{diag}(\Delta t_1, \ldots, \Delta t_n)$, $\delta = \mu$, $\Delta t' \Psi^{-1} \Delta t = T$, and his estimators are $\hat{\mu}$ and $\hat{\sigma}^2$ above. Hence, for $n \geq 2$,

$$\hat{\mu} \sim \mathcal{N}(\mu, \sigma^2/T), \qquad n\hat{\sigma}^2/\sigma^2 = \hat{Q}/\sigma^2 \sim \chi^2_{n-1},$$

$\hat{\mu}$ and $\hat{\sigma}^2$ are independent, $\hat{\mu}$ is unbiased and, the mean of a χ^2_{n-1} variable being $n-1$, $E[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2$: $\hat{\sigma}^2$ is biased downwards. The function also returns its unbiased version

$$s^2 = \frac{\hat{Q}}{n-1} = \frac{n}{n-1}\hat{\sigma}^2.$$

References

Kahle, W., Mercier, S. and Paroissin, C. (2016). *Degradation Processes in Reliability*. ISTE and Wiley, London. Definition 1.11, section 1.6.1 (equations [1.3], [1.14], [1.17] and [1.19]).

Whitmore, G. A. (1995). Estimating degradation by a Wiener diffusion process subject to measurement error. *Lifetime Data Analysis*, 1, 307-319. Section 3, equations (3.8a) to (3.10b).

See Also

gamma_mle(), ig_mle(), vg_mle(), meixner_mle()

Examples

```
# 1. One trajectory inspected every time unit up to 50 (mu = 0.5, sigma2 = 2.25)
t <- DegradSim::inspection_times("regular", n = 50, tmax = 50)$time
set.seed(1)
x <- DegradSim::sim_wiener(t, mu = 0.5, sigma = 1.5)
wiener_mle(x)

# The same trajectory given as two vectors: degradation levels, then times
wiener_mle(x[, "X1"], t)

# 2. One trajectory with irregular inspections, more and more frequent
t <- DegradSim::inspection_times("decreasing", n = 50, tmax = 50)$time
x <- DegradSim::sim_wiener(t, mu = 0.5, sigma = 1.5)
wiener_mle(x)

# 3. Ten trajectories with the same inspection times
t <- DegradSim::inspection_times("regular", n = 10, tmax = 50)$time
x <- DegradSim::sim_wiener(t, mu = 0.5, sigma = 1.5, n_traj = 10)
wiener_mle(x)

# 4. Ten trajectories, each with its own inspection times
plan <- DegradSim::inspection_times("increasing", n = 5:14, n_traj = 10, tmax = 50)
x <- DegradSim::sim_wiener(plan, mu = 0.5, sigma = 1.5)
wiener_mle(x)

# 5. Estimates with 95 % confidence intervals
wiener_mle(x, ci = TRUE)

# 6. Your own data: a list of data frames, one per trajectory,
#    with the times in the first column and the levels in the second
liste <- list(data.frame(t = c(0, 10, 20, 30), X = c(0, 4.2, 11.5, 13.9)),
```

```
              data.frame(t = c(0, 15, 30), X = c(0, 9.1, 12.8)))
wiener_mle(liste)

# sigma2 is the MLE and s2 its unbiased version: with few increments,
# here 3, they differ clearly
wiener_mle(c(0, 1.2, 0.7, 2.5), c(0, 1, 2, 3))
```

Index

∗ **package**
levyfit-package, 2

base::.Machine, *11*

gamma_mle, 5
gamma_mle(), *2*, *10*, *11*, *19*, *24*

ig_mle, 8
ig_mle(), *2*, *4*, *7*, *11*, *24*
increments_data, 11
increments_data(), *2*, *5*, *8*, *12*, *17*, *22*

levyfit *(*levyfit-package*)*, 2
levyfit-package, 2, *5*, *6*, *8*, *9*, *11–13*, *17*, *18*, *22*, *23*

meixner_mle, 12
meixner_mle(), *2–4*, *11*, *21*, *24*

set.seed(), *15*
stats::optim(), *3*, *15*, *18*
stats::optimHess(), *3*
stats::uniroot(), *3*, *7*

vg_mle, 17
vg_mle(), *2*, *11*, *16*, *24*

wiener_mle, 22
wiener_mle(), *2*, *7*, *10*, *11*, *19–21*